

KEYWORDS — parenting, discipline, child development

I. ACKNOWLEDGEMENTS

- Collaborators:
 - Professor Julie Ma
 - Professor Yoonsun Han
 - Dr. Kaitlin Ward
- Members of the *Global Families Project*
- IPUMS
- NICHD

II. THE BIG IDEAS

- i. *The Scope of the Issue*
- UNICEF considers children globally to be in crisis: political violence, climate change, and poverty affect many children (UNICEF, 2024a; 2024b).
 - In this time of crisis, parenting is one crucial area of focus.
 - According to UNICEF data, $\frac{1}{3}$ of children are subject to physical punishment (Ward et al., 2023); $\frac{2}{3}$ to some form of violent punishment (UNICEF, 2024a).
 - In a review of 50 years of literature on physical punishment, Gershoff & Grogan-Kaylor (2016) connect physical punishment to poor mental health outcomes, and other poor outcomes over the life course. Ward et al. (2023) examines the relationship of multiple forms of parental discipline to child outcomes.

ii. *Are We Similar? Are We Different?*



Figure 1: positive parenting

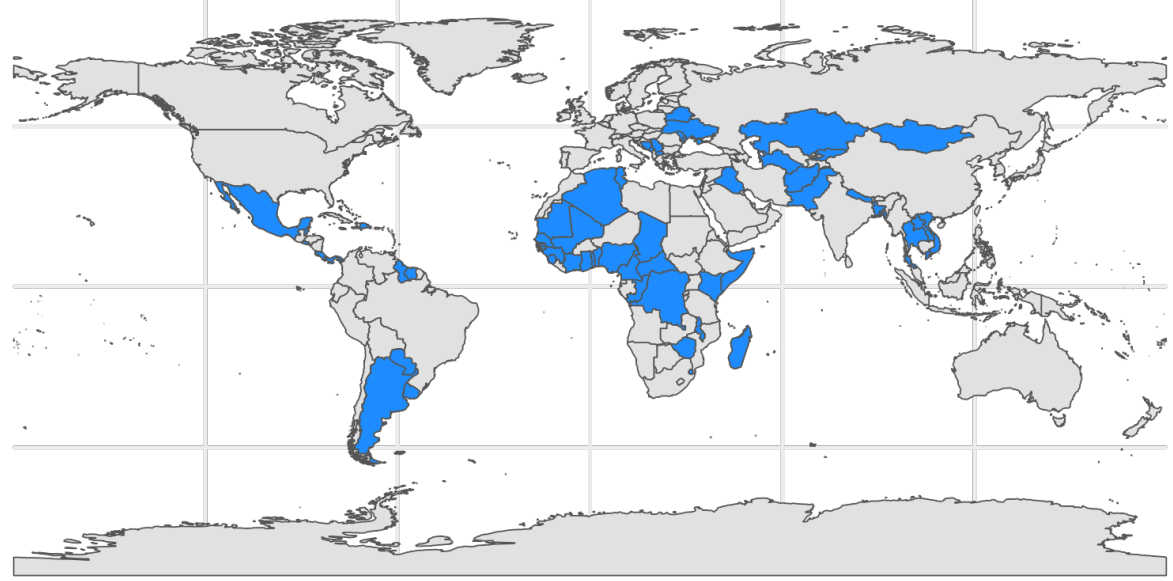


Figure 2: countries in MICS data

iii. *Both Universals and Variation Are Important*

"Despite the incredible diversity existing among and within human cultures, there are many phenomena that occur regularly in all known societies. These commonalities, or universals, while deriving in part from human nature, may also have specific social, cultural, and systemic sources. We need to develop a working understanding of these universals so that we might advance legitimate, empirically based human science set on creating knowledge that is politically relevant to fostering real solutions to the problems that complicate human co-existence in the Age of the Anthropocene." (Antweiler, 2016)

iv. *Highlighting Commonalities May Promote Intergroup Attitudes And Solidarity*

"Hanel et al. (2019) cite research suggesting that 'highlighting similarities between groups improves interpersonal and intergroup attitudes.' Others have argued that 'build[ing] bonds of commonality across our differences' might be an impetus for social transformation (Hunt-Hendrix & Taylor, 2024)."

v. *The Argument Of This Presentation*

- Families, parents and children experience an incredible diversity of circumstances.
- Parents engage in parenting behaviors with consequent diversity, though with some surprising similarities.
- The associations of parenting with child outcomes are surprisingly consistent.

III. THE STUDY

i. *Data*

MICS data collected by UNICEF from Low and Middle Income Countries (LMIC) around the world. $N = 224,632$ families; $N_{countries} = 60$

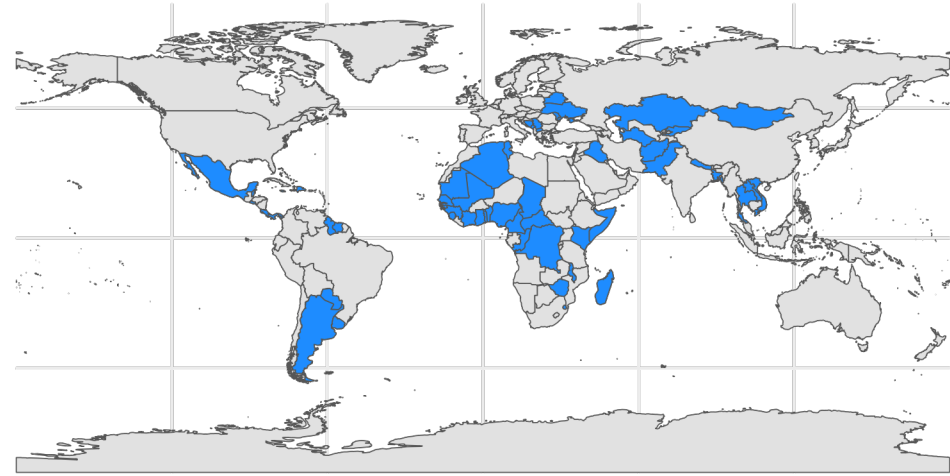


Figure 1: countries in UNICEF MICS Data

ii. *Sources of Variation In The Study of Parenting And Child Outcomes*

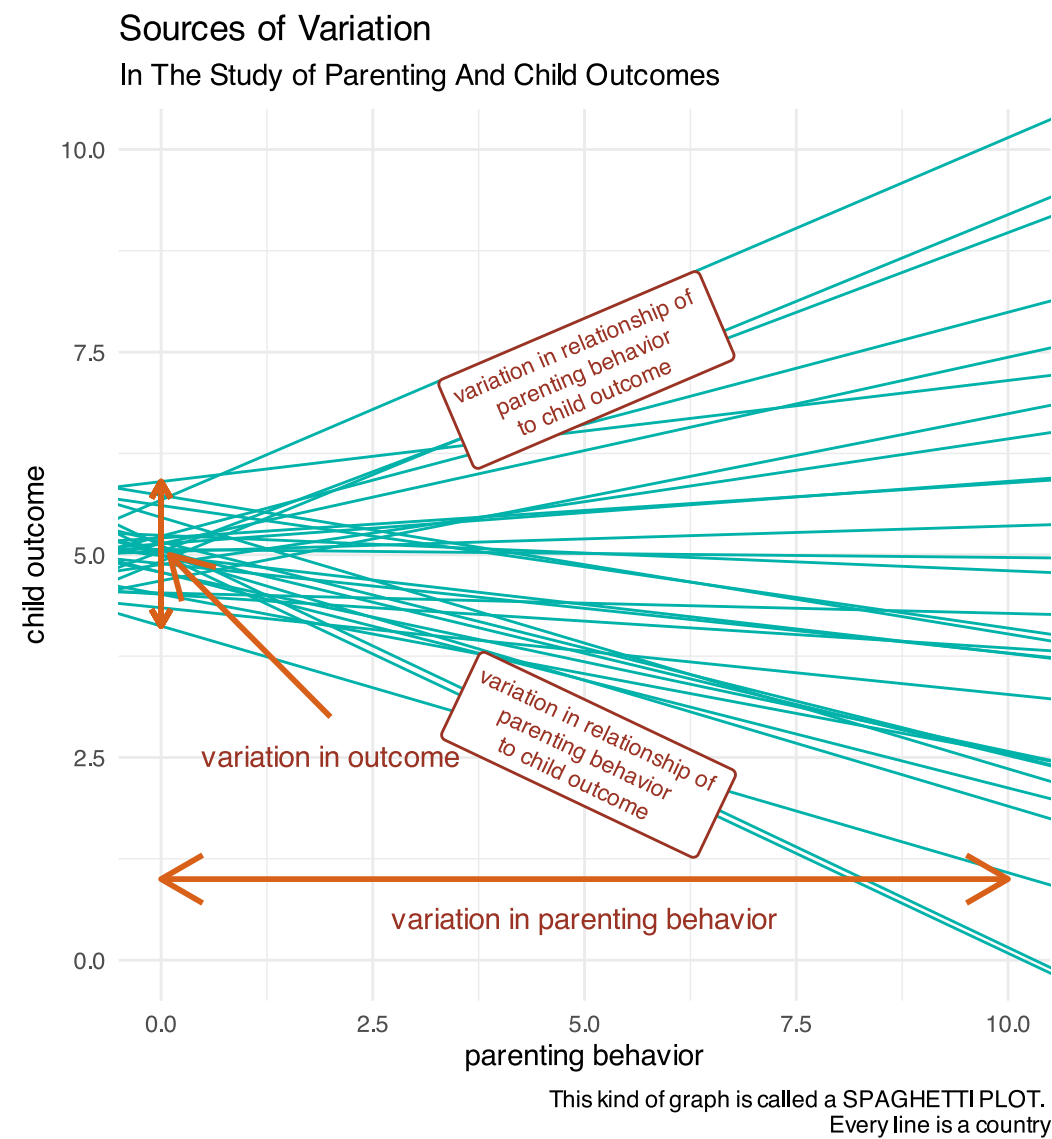


Figure 2: Sources of Variation In The Study of Parenting And Child Outcomes

iii. *Check-In*

Are we OK? Are there questions?

iv. *Sources of Variation (2)*

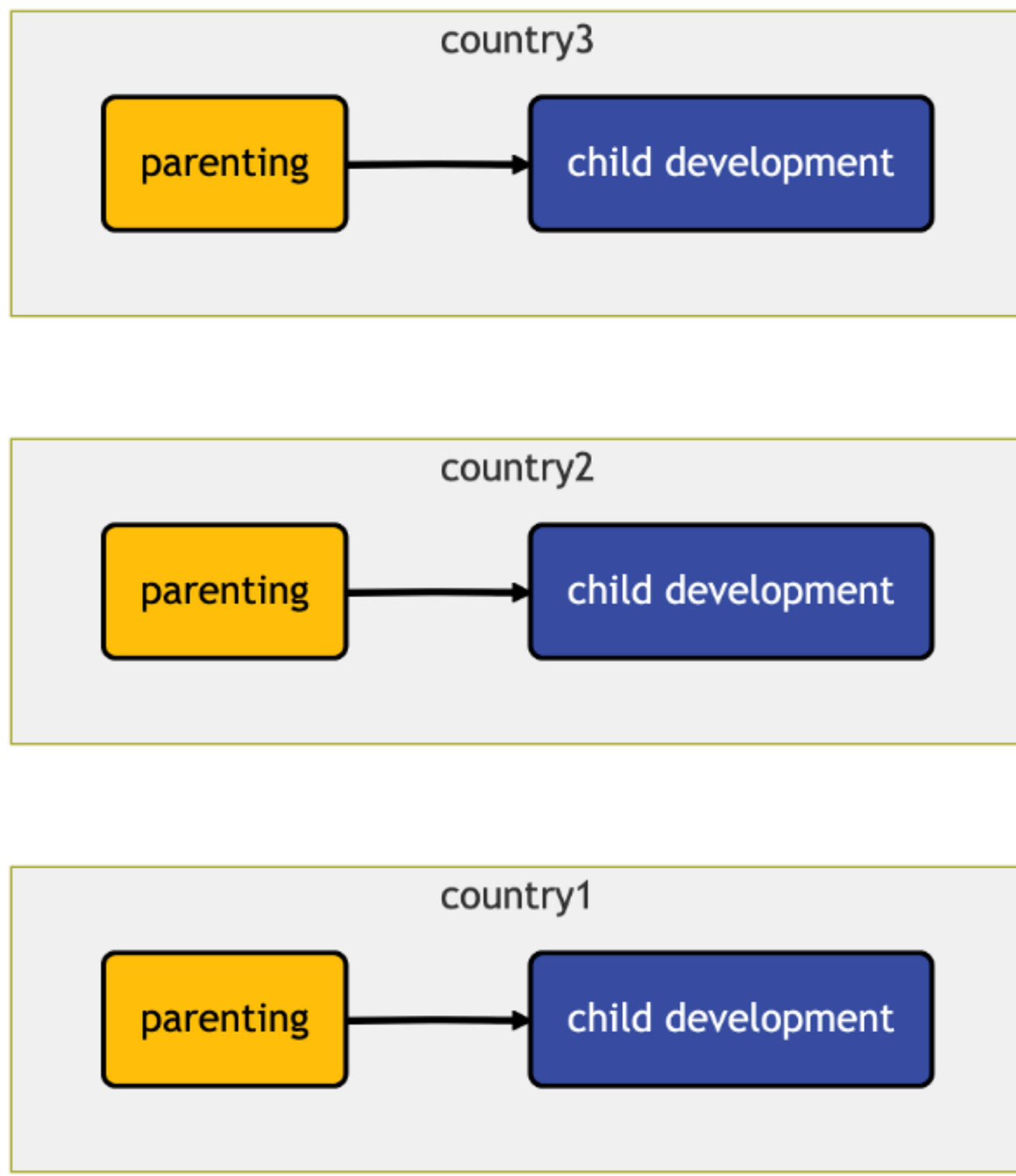


Figure 3: a visual version of the research question

v. *The Equation*

$$\text{outcome} = \beta_0 + \Sigma \beta \text{parenting} + \Sigma \beta \text{covariates} + u_{0j} + \Sigma u_{kj} \times \text{parenting}$$

Annotations:
- β_0 : baseline
- $\Sigma \beta \text{parenting}$: linear or dichotomous outcome
- $\Sigma \beta \text{covariates}$: covariates
- u_{0j} : country level variation in outcome
- $\Sigma u_{kj} \times \text{parenting}$: country level variation in associations

Figure 4: annotated equation

vi. *Parenting Behaviors Vary Somewhat Across Countries*

Verbal reasoning and shouting are the most common parental discipline behaviors toward young children.

Proportions of Parents Using Different Discipline Types
UNICEF MICS Data

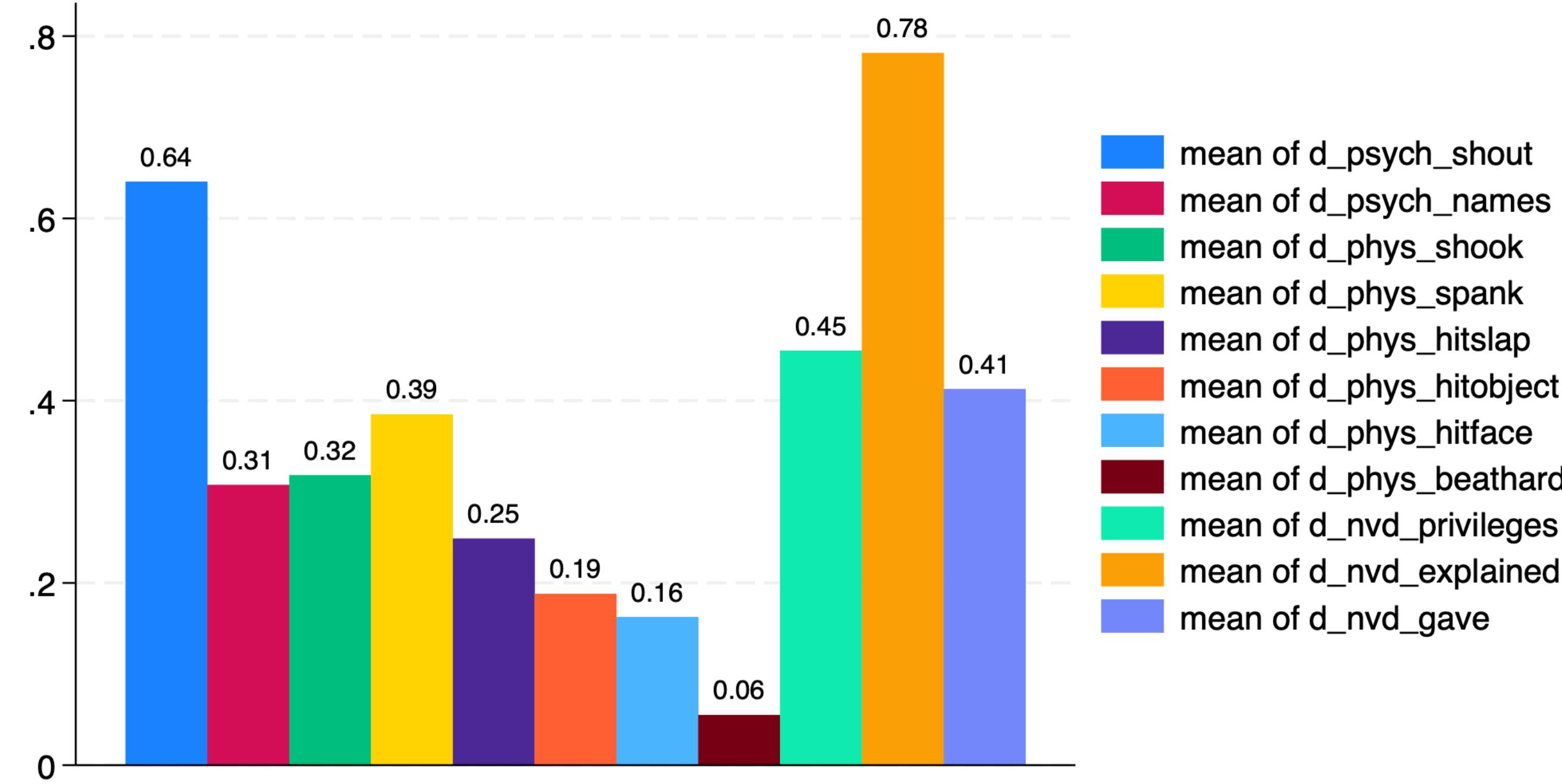


Figure 5: positive and negative parenting

vii. *The Maximal Model Approach*

- Hypothetically, one might imagine that there could be group level unobserved factors which affect many regression slopes—i.e. the relationship between multiple predictors (e.g. x_1, x_2, x_3 , etc.) and outcome variable y .
- Arguably, were one to ignore these unobserved factors in statistical estimation, they would show up either in an error term (e_i or u_{0i}), or in the regression coefficients (β). This could be seen as a form of statistical bias.
- The above could lead to a substantive mis-estimation of important effects. Thus, there is a conceptual argument for including as many random effects —i.e. random slopes—in a statistical model as possible (Barr et al., 2013; Frank, 2018).
- There is also a substantive reason: We might be interested in the *variation of all* the random slopes!

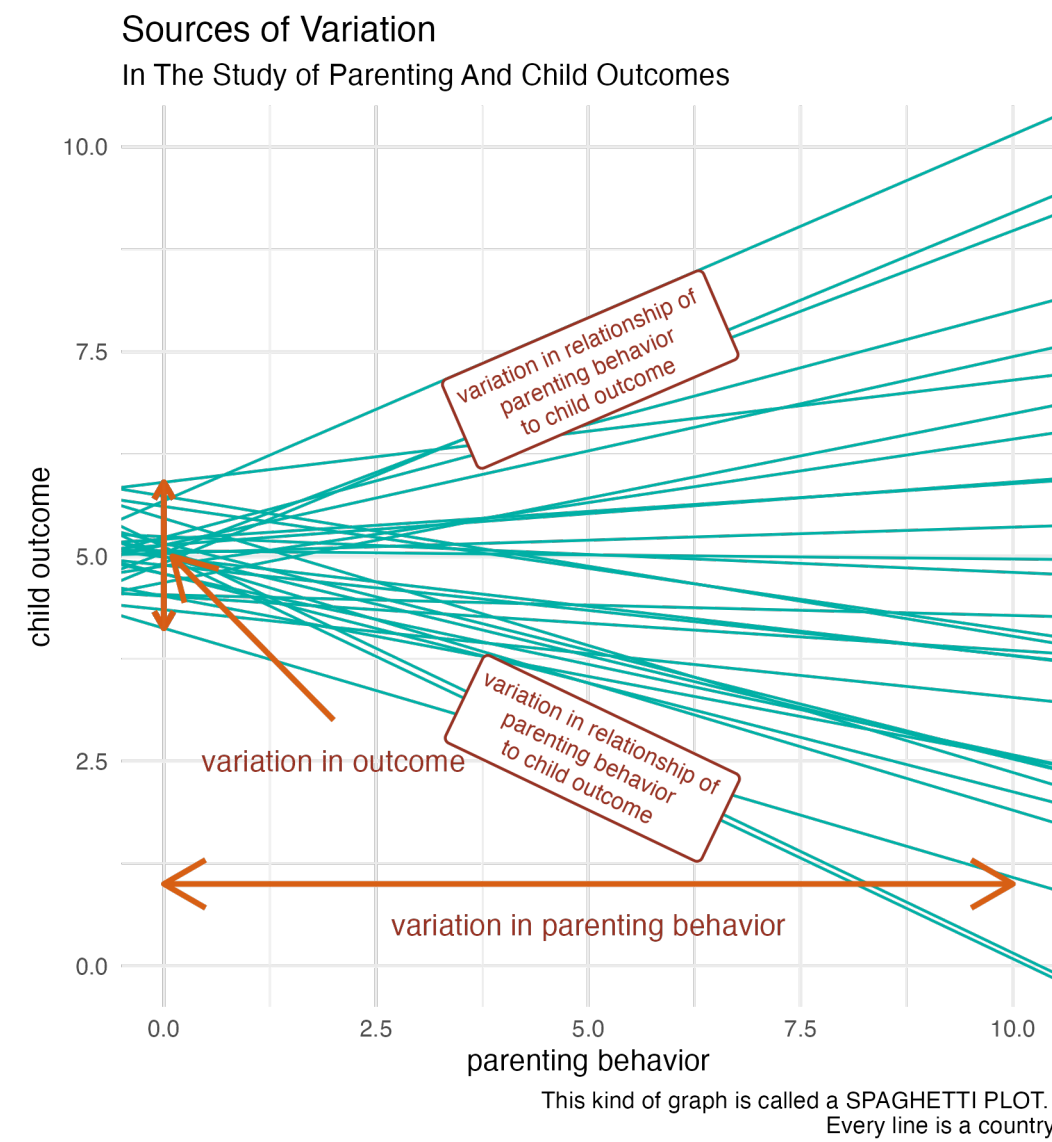


Figure 6: sources of variation

viii. *Relationship of Parenting And Child Outcomes*

Maximal Model of Parenting and Child Aggression Outcome

	aggress
child aggression	
Sex of randomly selected child	−0.023 **
Age of randomly selected child	0.000
Selected child shouted at	0.036 **
Selected child called dumb/lazy/names	0.054 **
Selected child shaken	0.039 **
Selected child spanked	0.023 **
Selected child hit/slapped on hand/arm/leg	0.033 **
Selected child hit with belt/brush/stick/etc	0.035 **
Selected child hit/slapped on face/head/ears	0.044 **
Selected child beat as hard as one could	0.026 **
Selected child's privileges taken away	0.007
Selected child - explained why behavior wrong	−0.014 *
Selected child given something else to do	0.004
Intercept	0.325 **
country	
var(d_psych_shout)	0.001
var(d_psych_names)	0.001
var(d_phys_shook)	0.001
var(d_phys_spank)	0.001
var(d_phys_hitslap)	0.001
var(d_phys_hitobject)	0.001
var(d_phys_hitface)	0.002
var(d_phys_beathard)	0.002
var(d_nvd_privileges)	0.001
var(d_nvd_explained)	0.001
var(d_nvd_gave)	0.001
var(_cons)	0.012
Residual	
var(e)	0.217
Number of observations	224632

** p<.01, * p<.05

IV. THE RELATIONSHIP OF PARENTING AND CHILD OUTCOMES SHOWS LITTLE VARIATION ACROSS COUNTRIES

i. *A Standard Spaghetti Plot*

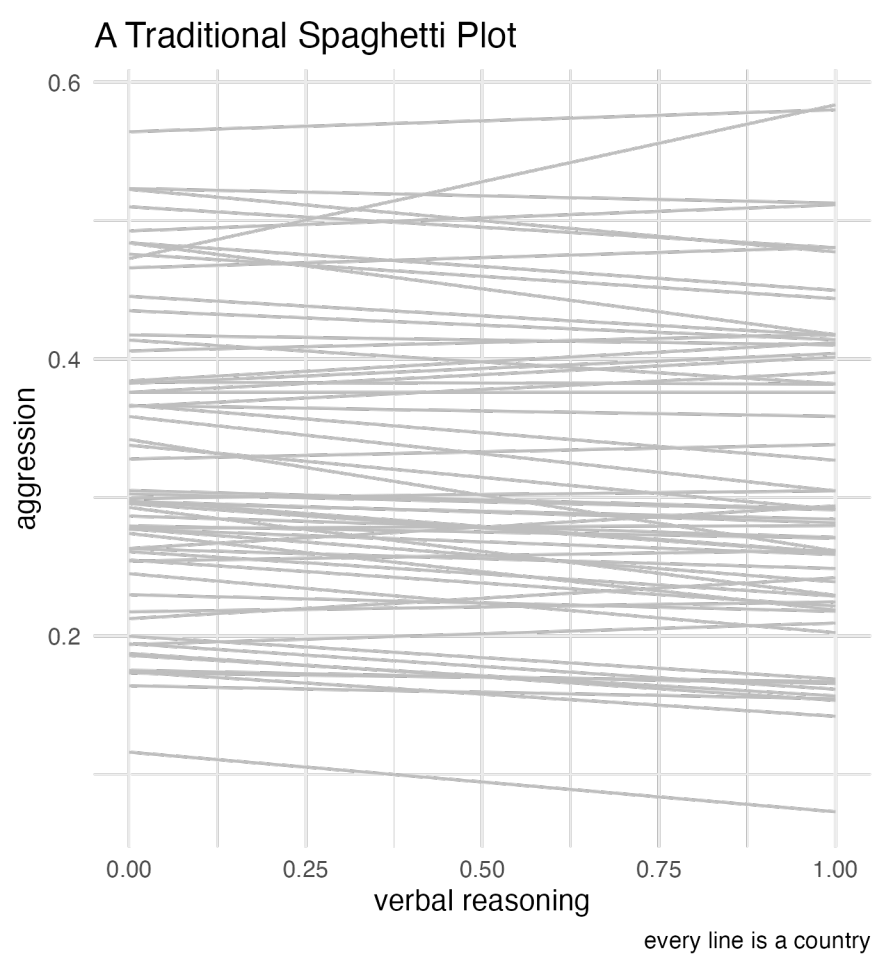


Figure 7: a standard 'spaghetti' plot

ii. *A Modified Spaghetti Plot*

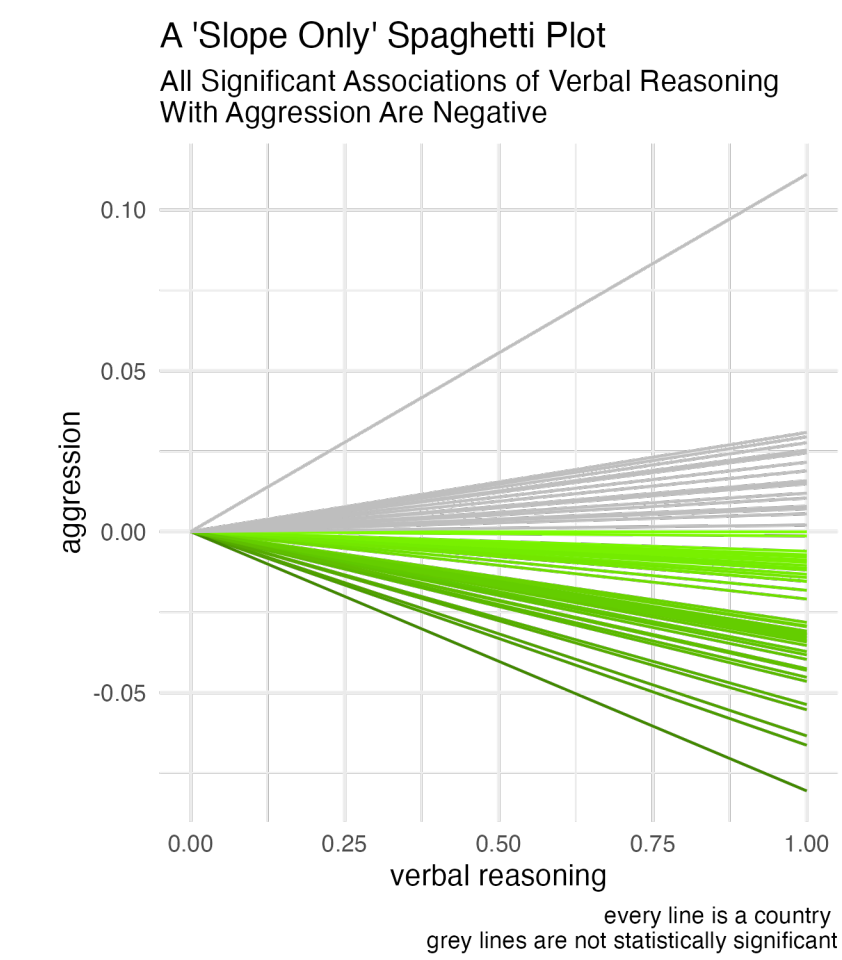


Figure 8: modified spaghetti plot

iii. Quantifying Diversity and Commonality in the Relationship of Parenting To Child Outcomes

- Regression slopes–when statistically significant–are all in the same direction, and of at least somewhat similar magnitude.
- The actual values of the var (u_{kj}) or range (u_{kj}) are small.

V. WRAPPING IT UP

i. To Summarize ...

- There is an incredible diversity across countries, especially LMIC's.
- However, the associations of different forms of parental discipline are surprisingly consistent.
 - Explaining to children the difference between right and wrong has consistently “good” outcomes in terms of aggression.
 - Spanking has consistently “bad” outcomes in terms of aggression.
 - Shouting has consistently “bad” outcomes in terms of aggression.
 - Name calling has consistently “bad” outcomes in terms of aggression.

ii. Attempting a Visual Summary...

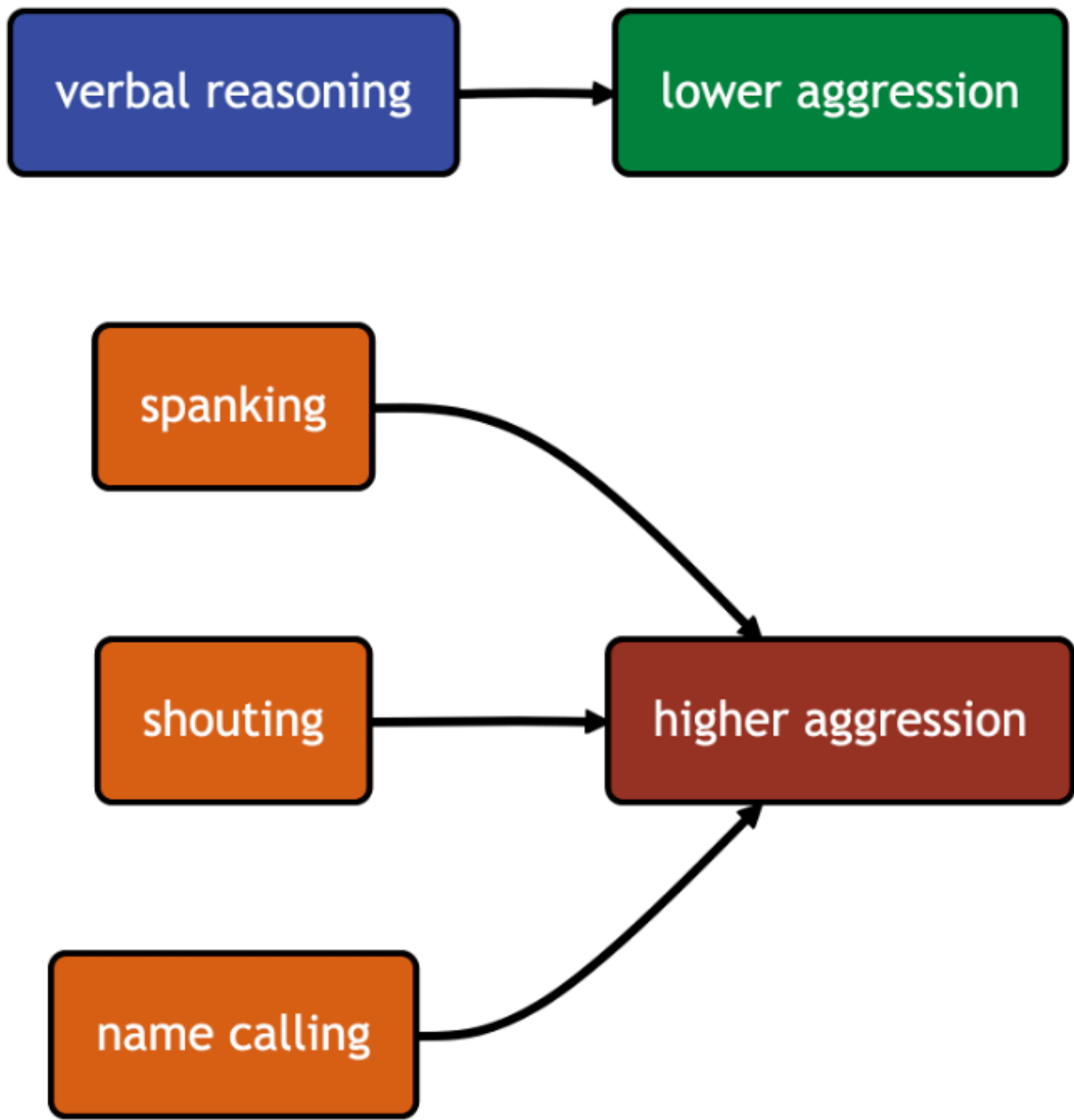


Figure 9: a visual summary

iii. Implications

“My conception of the universal is that of a universal enriched by all that is particular, a universal enriched by every particular: the deepening and coexistence of all particulars.” (Césaire, 1956)

- Associations of parental discipline with child outcomes are very consistent across countries.
- This may be helpful to international organizations (e.g. UNICEF, WHO) that are developing programs or interventions that are to be applied in multiple cultural settings.
- While cultural tailoring will always be necessary to some degree; child development research can inform the foundations of widely applicable interventions.

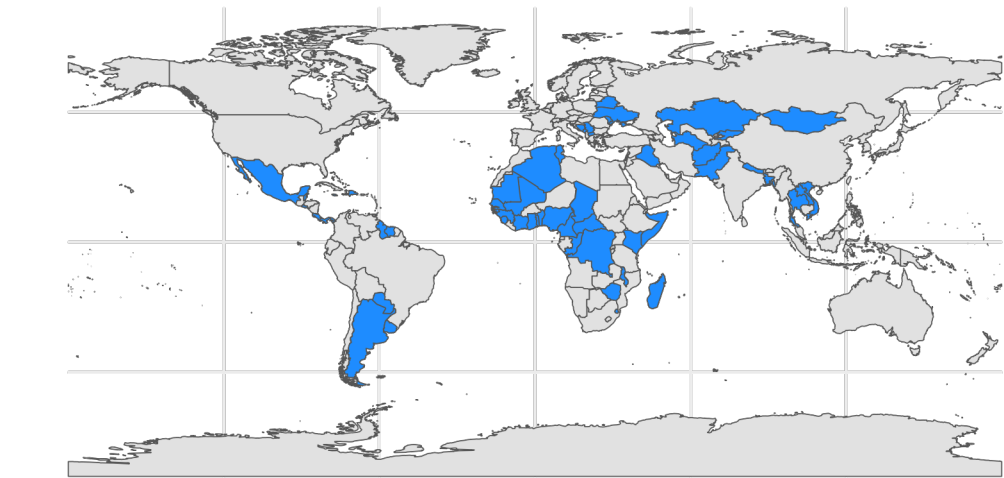


Figure 10: countries in MICS data

iv. Discussion and Questions

How might it be helpful to start thinking about possible universality, commonality, and similarity ...

... while keeping in mind ...

... our diversity, differences, and variation?

1) Other Questions?:



Figure 11: Global Families Project

BIBLIOGRAPHY

Antweiler, C. (2016). *Our common denominator: human universals revisited*. Berghahn.

Barr, D. J., Levy, R., Scheepers, C., & Tily, H. J. (2013). Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language*, 68(3), 255–278. <https://doi.org/10.1016/j.jml.2012.11.001>

Césaire, A. (1956). Letter to Maurice Thorez. *Présence Africaine*.

Frank, M. (2018). *Mixed effects models: Is it time to go Bayesian by default?*. <http://babieslearninglanguage.blogspot.com/2018/02/mixed-effects-models-is-it-time-to-go.html>

Gershoff, E. T., & Grogan-Kaylor, A. (2016). Spanking and child outcomes: Old controversies and new meta-analyses. *Journal of Family Psychology*, 30(4), 453–469. <https://doi.org/10.1037/fam0000191>

Hanel, P. H. P., Maio, G. R., & Manstead, A. S. R. (2019). A new way to look at the data: Similarities between groups of people are large and important. *Journal of Personality and Social Psychology*, 116(4), 541–562. <https://doi.org/10.1037/pspi0000154>

Hunt-Hendrix, L., & Taylor, A. (2024). *Solidarity: the past, present, and future of a world-changing idea*. Pantheon Books.,

UNICEF. (2024a). *FAST FACTS: Violence against children widespread, affecting millions globally*. <https://www.unicef.org/press-releases/fast-facts-violence-against-children-widespread-affecting-millions-globally>

UNICEF. (2024b). *'Not the new normal' – 2024 'one of the worst years in UNICEF's history' for children in conflict*. <https://www.unicef.org/press-releases/not-new-normal-2024-one-worst-years-unicefs-history-children-conflict>

Ward, K. P., Grogan-Kaylor, A. C., Ma, J., Pace, G., & Lee, S. J. (2023). Associations Between 11 Parental Discipline Behaviors and Child Outcomes Across 60 Countries. *BMJ Open*. <https://doi.org/10.1136/bmjopen-2021-058439>